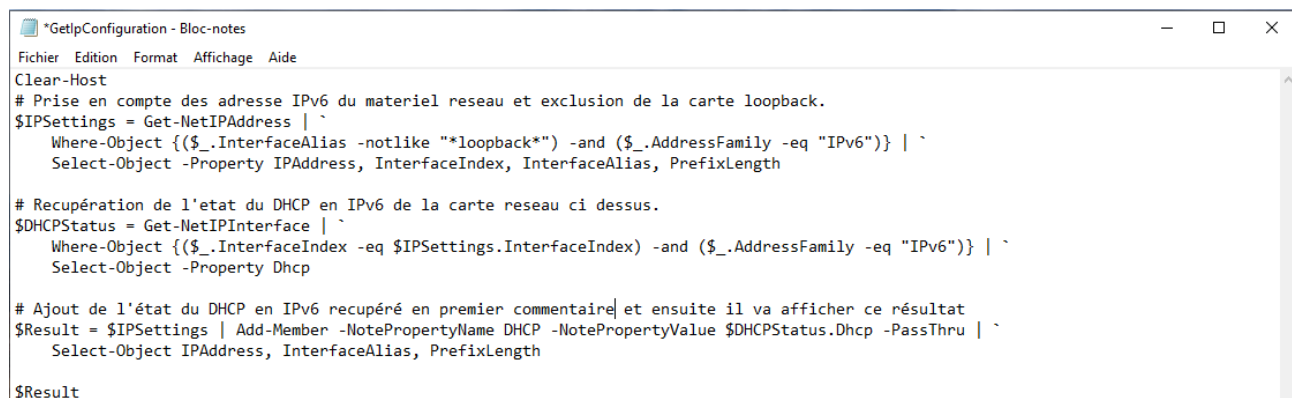


Fichier réponse de la Mise En Situation Professionnelle

1. Adaptation de script

Intervention 1

Question 1 :



```
*GetIpConfiguration - Bloc-notes
Fichier Edition Format Affichage Aide
Clear-Host
# Prise en compte des adresse IPv6 du materiel reseau et exclusion de la carte loopback.
$IPSettings = Get-NetIPAddress | `
    Where-Object {($_.InterfaceAlias -notlike '*loopback*') -and ($_.AddressFamily -eq "IPv6")} | `
    Select-Object -Property IPAddress, InterfaceIndex, InterfaceAlias, PrefixLength

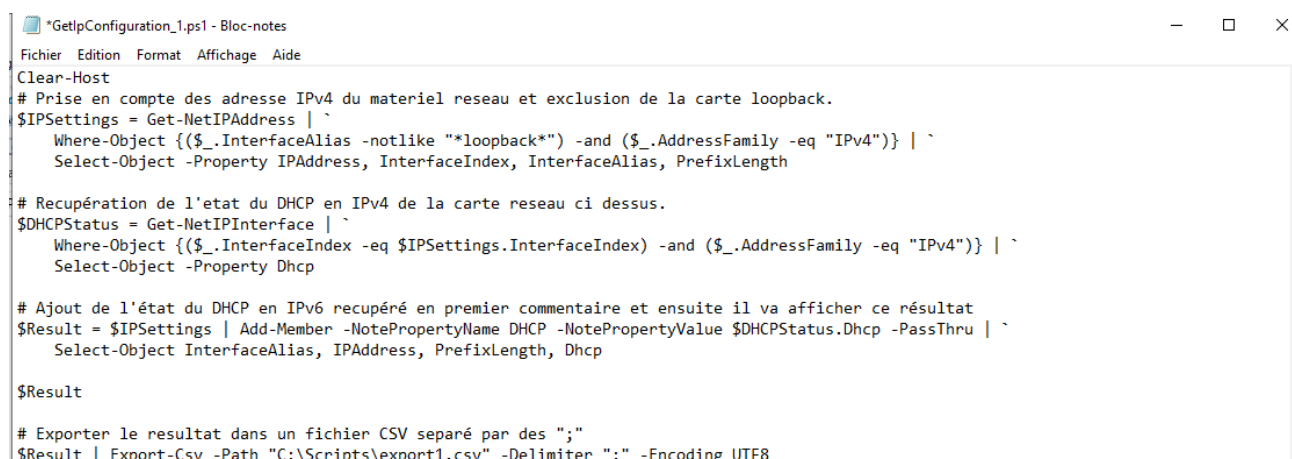
# Recuperation de l'etat du DHCP en IPv6 de la carte reseau ci dessus.
$DHCPStatus = Get-NetIPInterface | `
    Where-Object {($_.InterfaceIndex -eq $IPSettings.InterfaceIndex) -and ($_.AddressFamily -eq "IPv6")} | `
    Select-Object -Property Dhcp

# Ajout de l'etat du DHCP en IPv6 recupere en premier commentaire et ensuite il va afficher ce resultat
$Result = $IPSettings | Add-Member -NotePropertyName DHCP -NotePropertyValue $DHCPStatus.Dhcp -PassThru | `
    Select-Object IPAddress, InterfaceAlias, PrefixLength

$Result
```

Intervention 2

Question 2 :



```
*GetIpConfiguration_1.ps1 - Bloc-notes
Fichier Edition Format Affichage Aide
Clear-Host
# Prise en compte des adresse IPv4 du materiel reseau et exclusion de la carte loopback.
$IPSettings = Get-NetIPAddress | `
    Where-Object {($_.InterfaceAlias -notlike '*loopback*') -and ($_.AddressFamily -eq "IPv4")} | `
    Select-Object -Property IPAddress, InterfaceIndex, InterfaceAlias, PrefixLength

# Recuperation de l'etat du DHCP en IPv4 de la carte reseau ci dessus.
$DHCPStatus = Get-NetIPInterface | `
    Where-Object {($_.InterfaceIndex -eq $IPSettings.InterfaceIndex) -and ($_.AddressFamily -eq "IPv4")} | `
    Select-Object -Property Dhcp

# Ajout de l'etat du DHCP en IPv4 recupere en premier commentaire et ensuite il va afficher ce resultat
$Result = $IPSettings | Add-Member -NotePropertyName DHCP -NotePropertyValue $DHCPStatus.Dhcp -PassThru | `
    Select-Object InterfaceAlias, IPAddress, PrefixLength, Dhcp

$Result

# Exporter le resultat dans un fichier CSV separe par des ';'
$Result | Export-Csv -Path "C:\Scripts\export1.csv" -Delimiter ";" -Encoding UTF8
```

Windows PowerShell ISE

Fichier Modifier Afficher Outils Débugger Composants additionnels Aide

Sans titre1.ps1 X

```

1 Clear-Host
2 # Prise en compte des adresse IPv4 du materiel reseau et exclusion de la carte loopback.
3 $IPSettings = Get-NetIPAddress | `
4     Where-Object {($_.InterfaceAlias -notlike "*loopback*") -and ($_.AddressFamily -eq "IPv4")} | `
5     Select-Object -Property IPAddress, InterfaceIndex, InterfaceAlias, PrefixLength
6
7 # Recuperation de l'etat du DHCP en IPv4 de la carte reseau ci dessus.
8 $DHCPStatus = Get-NetIPInterface | `
9     Where-Object {($_.InterfaceIndex -eq $IPSettings.InterfaceIndex) -and ($_.AddressFamily -eq "IPv4")} | `
10    Select-Object -Property Dhcp
11
12 # Ajout de l'etat du DHCP en IPv6 recupere en premier commentaire et ensuite il va afficher ce resultat
13 $Result = $IPSettings | Add-Member -NotePropertyName DHCP -NotePropertyValue $DHCPStatus.Dhcp -PassThru | `
14    Select-Object InterfaceAlias, IPAddress, PrefixLength, Dhcp
15
16 $Result
17
18 # Exporter le resultat dans un fichier CSV separe par des ";"
19 $Result | Export-Csv -Path "C:\Scripts\export1.csv" -Delimiter ";" -Encoding UTF8

```

InterfaceAlias	IPAddress	PrefixLength	DHCP
Ethernet 2	192.168.1.30	24	
Ethernet	169.254.151.221	16	

export1 - Bloc-notes

Fichier Edition Format Affichage Aide

```

#TYPE Selected.System.Management.Automation.PSCustomObject
"InterfaceAlias";"IPAddress";"PrefixLength";"DHCP"
"Ethernet 2";"192.168.1.30";"24";
"Ethernet";"169.254.151.221";"16";

```

pts

Nom	Modifié le	Type
export1	08/07/2026 09:42	Fich
GetIpConfiguration	08/07/2026 09:31	Script
GetIpConfiguration_1.ps1	08/07/2026 09:37	Script
GetIpConfiguration_2.ps1	08/07/2026 09:31	Script

Question 3 :

GetIpConfiguration_2.ps1 - Bloc-notes

Fichier Edition Format Affichage Aide

```

Clear-Host
# Prise en compte des adresse IPv4 du materiel reseau et exclusion de la carte loopback.
$IPSettings = Get-NetIPAddress | `
    Where-Object {($_.InterfaceAlias -notlike "*loopback*") -and ($_.AddressFamily -eq "IPv4")} | `
    Select-Object -Property IPAddress, InterfaceIndex, InterfaceAlias, PrefixLength
# Recuperation de l'etat du DHCP en IPv4 de la carte reseau ci dessus.
$DHCPStatus = Get-NetIPInterface | `
    Where-Object {($_.InterfaceIndex -eq $IPSettings.InterfaceIndex) -and ($_.AddressFamily -eq "IPv4")} | `
    Select-Object -Property Dhcp
# Ajout de l'etat du DHCP en IPv6 recupere en premier commentaire et ensuite il va afficher ce resultat
$Result = $IPSettings | Add-Member -NotePropertyName DHCP -NotePropertyValue $DHCPStatus.Dhcp -PassThru | `
    Select-Object InterfaceAlias, IPAddress, PrefixLength, Dhcp
$Result
# Exporter le resultat dans un fichier CSV separe par des ";"
$Result | Export-Csv -Path "C:\Scripts\export2.csv" -delimiter ";" -Encoding UTF8 |
Write-Host "Interface type : $InterfaceAlias"
Write-Host "Adresse IP : $IPAddressCIDR"
Write-Host "Etat du DHCP : $Dhcp"

```

Adresse IP :
CIDR :
Etat du DHCP :
IPAddress InterfaceAlias PrefixLength DHCP
192.168.1.30 Ethernet 2 24
169.254.151.221 Ethernet 16

PS C:\Users\wilders>

export2 - Bloc-notes

Fichier Edition Format Affichage Aide

```

#TYPE Selected.System.Management.Automation.PSCustomObject
"IPAddress";"InterfaceAlias";"PrefixLength";"DHCP"
"192.168.1.30";"Ethernet 2";"24";
"169.254.151.221";"Ethernet";"16";

```

Ce PC > Disque local (C:) > Scripts

Nom
export1
export2
GetIpConfiguration
GetIpConfiguration_1.ps1
GetIpConfiguration_2.ps1

2. Active Directory

Intervention 1

Question 4 :

Administrator: Windows PowerShell

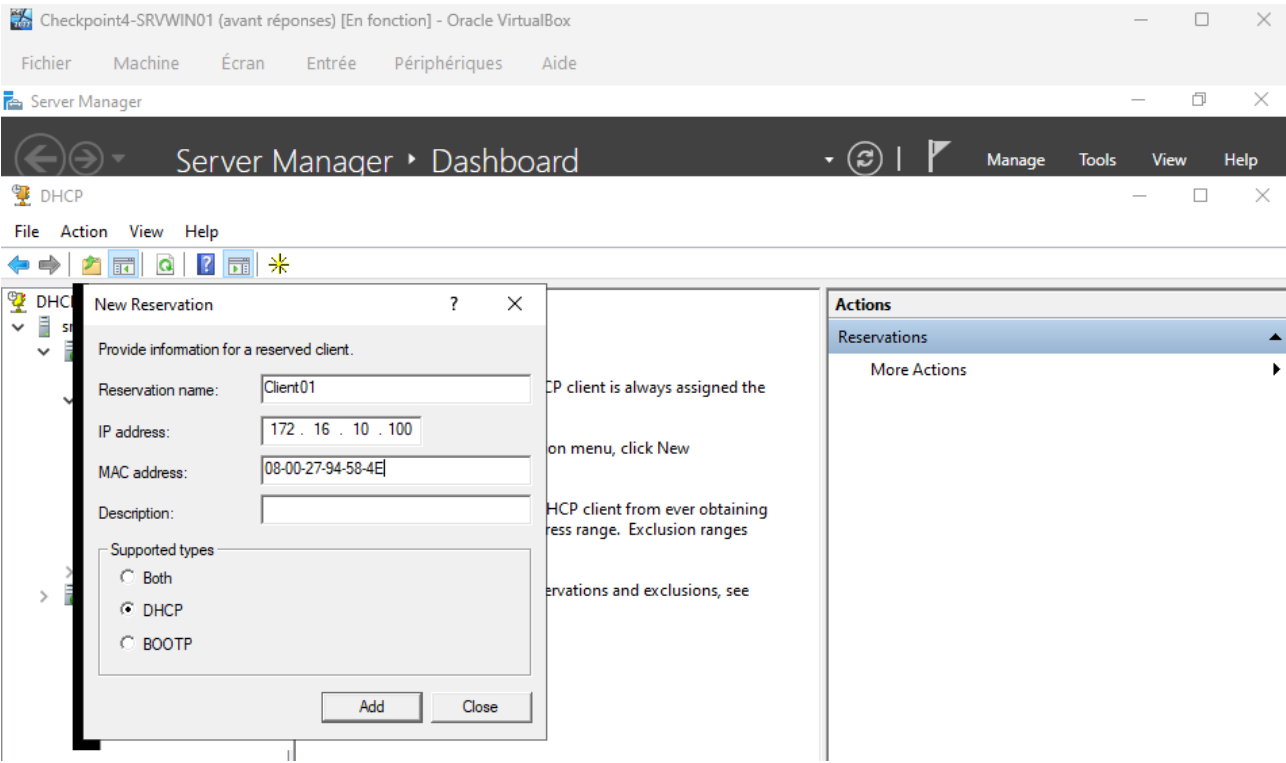
```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

Host Name . . . . . : SRVWIN01
Primary Dns Suffix . . . . . : lab.lan
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : lab.lan


Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-A7-98-C7
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::4517:688a:f4f3:e574%7(Preferred)
IPv4 Address. . . . . : 172.16.10.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-DE-A3-55-08-00-27-A7-98-C7
DNS Servers . . . . . : ::1
                        127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled
```



```

PS C:\Users\wilder> ipconfig /all
192.168.1.254

Configuration IP de Windows

Nom de l'hôte . . . . . : CLIENT01
Suffixe DNS principal . . . . . : lab.lan
Type de noeud . . . . . : Hybride
Routage IP activé . . . . . : Non
Proxy WINS activé . . . . . : Non
Liste de recherche du suffixe DNS.: lab.lan

Carte Ethernet Ethernet :

Suffixe DNS propre à la connexion. . . : lab.lan
Description. . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Adresse physique . . . . . : 08-00-27-94-58-4E
DHCP activé. . . . . : Oui
Configuration automatique activée. . . : Oui
Adresse IPv6 de liaison locale. . . . : fe80::fd1a:602a:c93d:21bf%10(préfér  )
Adresse IPv4. . . . . : 172.16.10.100(pr  f  r  )
Masque de sous-r  seau. . . . . : 255.255.255.0
Bail obtenu. . . . . : mercredi 8 juillet 2026 10:15:24
Bail expirant. . . . . : jeudi 16 juillet 2026 10:15:28
Passerelle par d  faut. . . . . :
Serveur DHCP . . . . . : 172.16.10.10
IAID DHCPv6 . . . . . : 151519271
DUID de client DHCPv6. . . . . : 00-01-00-01-31-DE-A1-76-08-00-27-94-58-4E
Serveurs DNS. . . . . : 172.16.10.10
NetBIOS sur Tcpip. . . . . : Activ  

```

Intervention 2

Question 5 :

The screenshot shows the 'Active Directory Users and Computers' window for the 'lab.lan' domain. The left pane shows the tree structure with 'LabUsers' expanded, and 'Gestion' selected. The right pane displays a list of objects:

Name	Type	Description
aibrahim	User	
akumar	User	
cwei	User	
fdiallo	User	
fivanovich	User	
GL-Wallpaper-Gestion	Security Group - Global	
GrpComptables	Security Group - Domain Local	
GrpFinanciers	Security Group - Domain Local	
irossi	User	
jsmith	User	
jtaylor	User	
sbrown	User	

Checkpoint4-SRVWIN01 (avant réponses) [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers [SRVWIN01.lab.lan]

- Saved Queries
- lab.lan
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - LabComputers
 - LabSecurity
 - LabUsers
 - Direction
 - Gestion
 - Informatique
 - Marketing
 - Production
 - Managed Service Accounts
 - Users

Name	Type	Description
aibrahim	User	
akumar	User	
cwei	User	
fdiallo	User	
fivanovich	User	
GrpComptables	Security Group - Domain Local	
GrpFinanciers	Security Group - Domain Local	
Grp-Wallpaper-Gestion	Security Group - Global	
irossi	User	
jsmith	User	
jtaylor	User	
sbrown	User	

Checkpoint4-SRVWIN01 (avant réponses) [En fonction] - Oracle VirtualBox

Fichier Machine Écran Entrée Périphériques Aide

Group Policy Management

File Action View Window Help

Group Policy Management

- Forest: lab.lan
 - Domains
 - lab.lan
 - Default Domain Policy
 - Domain Controllers
 - LabComputers
 - LabSecurity
 - LabUsers
 - USER-Interface-Wallpaper-Direction
 - USER-Interface-Wallpaper-Informatique
 - USER-Interface-Wallpaper-Marketing
 - USER-Interface-Wallpaper-Production
 - USER-Reseau-MapLecteurCommun
 - USER-Reseau-MapLecteurIndividuel
 - Direction
 - Gestion
 - USER-INTERFACE-WALLPAPER-GESTION
 - Informatique
 - Marketing
 - Production
 - Group Policy Objects
 - WMI Filters
 - Starter GPOs
 - Sites
 - Group Policy Modeling
 - Group Policy Results

USER-INTERFACE-WALLPAPER-GESTION

Scope Details Settings Delegation

Links

Display links in this location: lab.lan

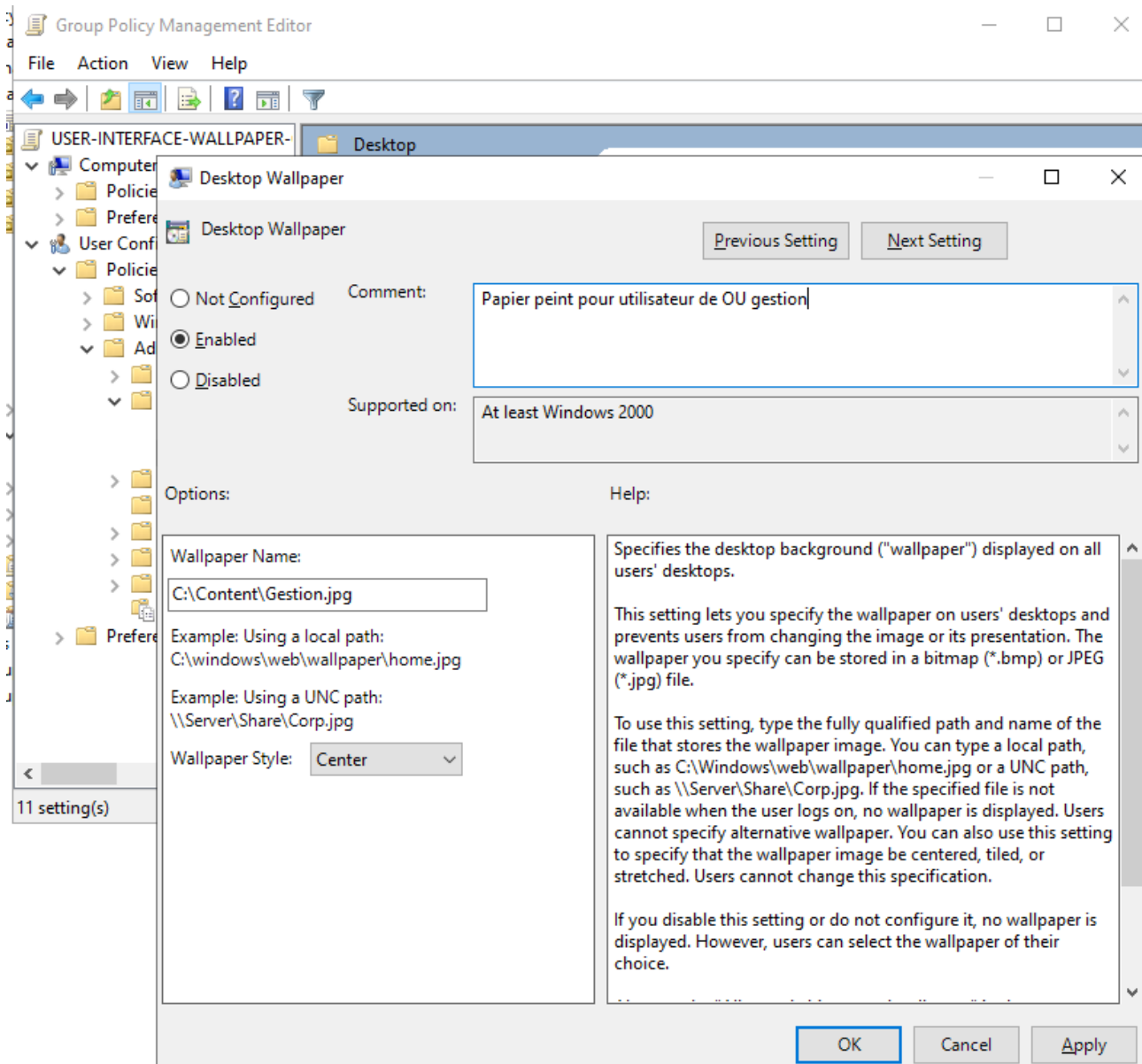
The following sites, domains, and OUs are linked to this GPO:

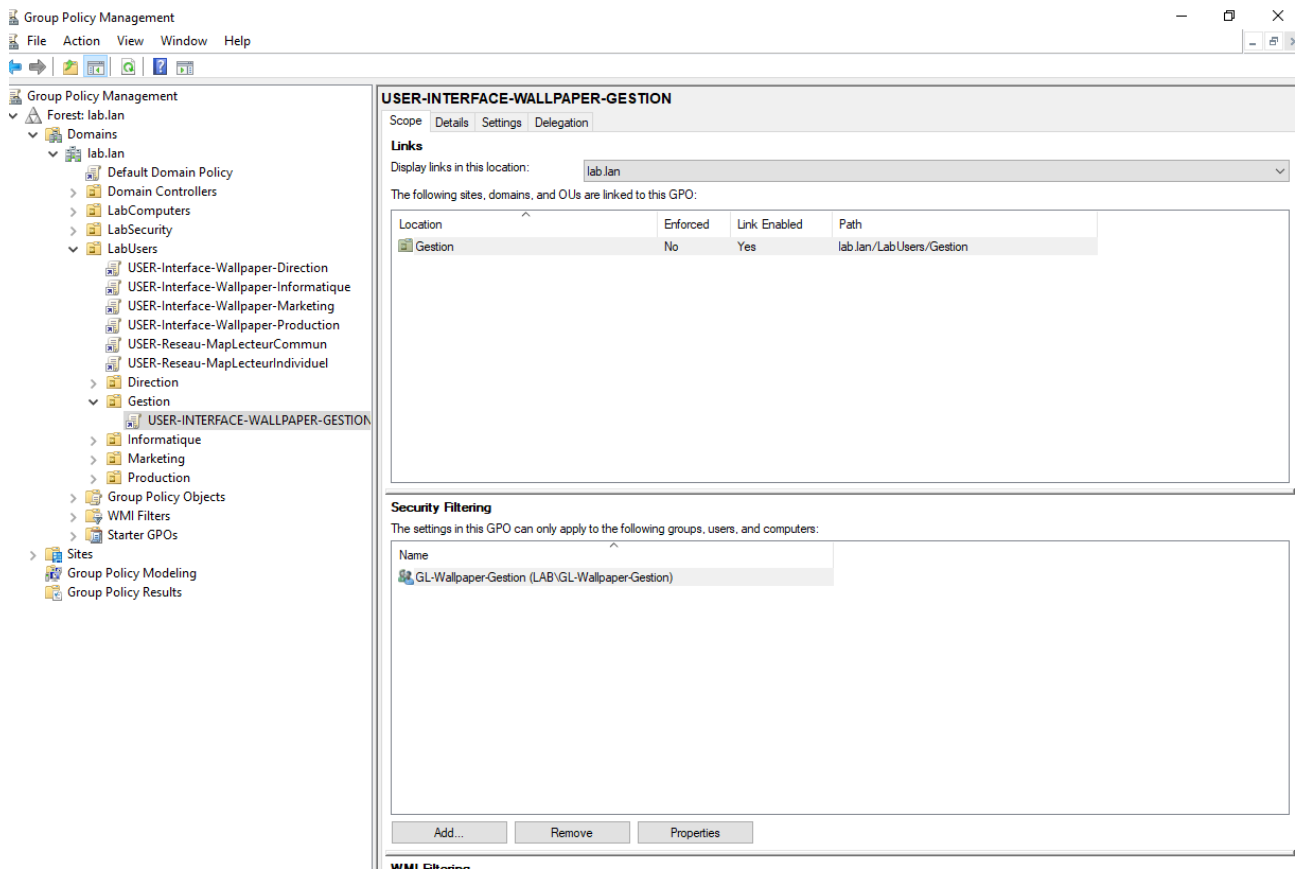
Location	Enforced	Link Enabled	Path
Gestion	No	Yes	lab.lan/LabUsers/Gestion

Security Filtering

The settings in this GPO can only apply to the following groups, users, and computers:

Name
Authenticated Users





Intervention 3

Question 6 :

3. Serveur Linux

Intervention 1

Question 7 :

```
GNU nano 5.4 /etc/ssh/sshd_config *
# Authentication:
#LoginGraceTime 2m
PermitRootLogin no_
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none
```

```
GNU nano 5.4 /etc/ssh/sshd_config *
#LogLevel INFO

# Authentication:
#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no_
#PermitEmptyPasswords no
```



```
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
```

```
GNU nano 5.4 /etc/ssh/sshd_config *
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 22504

#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
```

Intervention 2

Question 8 :

4. Téléphonie

Intervention 1

Question 9 :

Intervention 2

Question 10 :